

Assignment 1 Queryquest. 240352_Arnabhoot.

Part I-

1) Traditional file system- saving data in separate files everywhere, so things can get messy and unorganized, Very difficult to collaborate for multiple users, and issues of data redundancy, poor security, inconsistencies and long data lookup times. DBMS- software system that stores and manages data in a structured way. Allows multiple users to access and update the same data efficiently. It improves data consistency, security, integrity, backup, and allows more efficient operations. Uses APIs.

2) Weak entity- double rectangle, Strong entity- single rectangle. The primary key of a weak entity is formed using a relationship with the primary key of the strong entity. Its known as identifying relationship.

3) Multivalued attribute as a student can have multiple phone nos.

4) A super key is any set of attributes that can uniquely identify a student in the table. Super Keys that are not Candidate Keys: (roll no, name), (email, branch). A candidate key is the smallest possible super key, meaning no attribute can be removed from it and still uniquely identify a student. Roll no and name are all possible candidate keys assuming they are both unique for every student.

5) The foreign key constraint is violated. It protects against invalid or unlinked records and ensures that every enrollment is linked to a valid student. The insert operation should be rejected by the DBMS.

6) An additional junction (linking) table must be created. The primary key of this table usually consists of the primary keys of both participating entities combined. The relationship cannot be stored as a column in one table because one record can be related to many records in the other table, which would lead to multiple values in a single column and violate relational database rules.

Part II: Long Answer Question

Part A – Entity and Relationship Identification

Entities and Key Attributes: Customer (customer_id, name, phone), Restaurant (restaurant_id, restaurant_name, address) , MenuItem (item_id, item_name, price) ,Order (order_id, order_date)

The relationship between Customer and Order is 1:N because one customer can place multiple orders, but each order belongs to only one customer.

The relationship between Restaurant and Order is 1:N because one restaurant can receive many orders, while each order is associated with exactly one restaurant.

The relationship between Restaurant and MenuItem is 1:N because a restaurant can offer multiple menu items, but each menu item belongs to one restaurant.

The relationship between Order and MenuItem is M:N because an order can contain multiple menu items, and a menu item can appear in multiple orders.

Part B – Attribute Classification

A multivalued attribute is phone number because a customer may have more than one phone number.

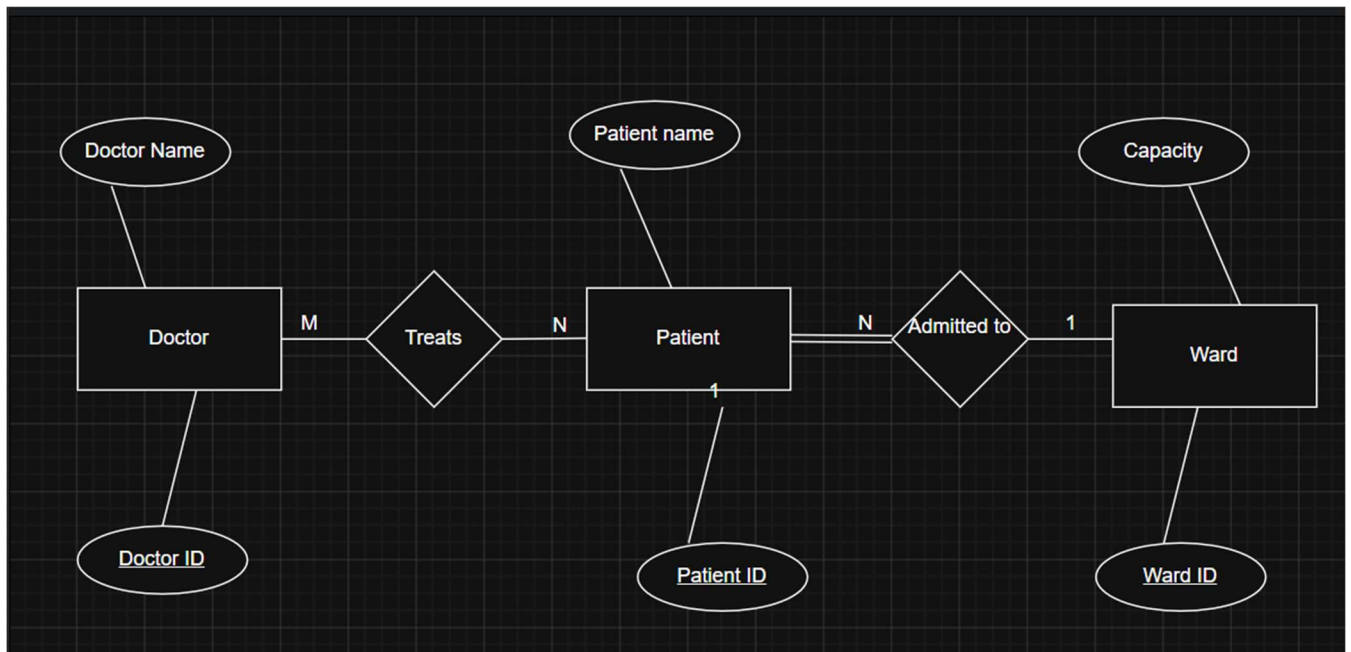
A composite attribute is address because it can be divided into smaller parts such as street, city, state, and pincode.

Part C – Relational Schema Design

Order_Menuitem(order_id, item_id, quantity).

The primary key is (order_id, item_id). The foreign keys are order_id, which references Order(order_id), and item_id, which references MenuItem(item_id). Additional attribute- number of units of a menu item included in an order.

Part III: ER Diagram Design



Question 2: Design Justification

The Doctor–Patient relationship is M:N because a doctor can treat many patients and a patient can be treated by multiple doctors during their stay. Associative entity created is called Treats (or Doctor_Patient). The composite primary key of this table would be (doctor_id, patient_id), since both together uniquely identify each treatment relationship.

In the Patient–Ward relationship, the participation of patient is total because every patient admitted in the hospital must be assigned to exactly one ward at a time. However, Ward has partial participation because a ward can exist even if it currently has no patients. In the Doctor–Patient relationship, both Doctor and Patient have partial participation since doctors may not always have assigned patients at every moment and patients are not necessarily being treated continuously.

I modelled address as a composite attribute because it can naturally be broken into smaller parts like street, city, state, and pincode.